

Using common series

Basic skills

Q1. a) $\sum_{r=1}^5 r = \frac{5 \times (5+1)}{2} = 15$ $\leftarrow$ use $\sum_{r=1}^n r = \frac{n(n+1)}{2}$

b) $\sum_{r=1}^7 r^2 = \frac{7 \times 8 \times 15}{6} = 140$ $\leftarrow$ use $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$

c) $\sum_{r=1}^{11} r^3 = \frac{11^2 \times 12^2}{4} = 4356$ $\leftarrow$ use $\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4}$

d) $1^2 + \dots + 18^2 = \sum_{r=1}^{18} r^2 = \frac{18 \times 19 \times 37}{6} = 2109$

e) $1^3 + \dots + 10^3 = \sum_{r=1}^{10} r^3 = \frac{10^2 \times 11^2}{4} = 3025$

f) $2 \sum_{r=1}^7 r^2 - \sum_{r=1}^5 r = 2 \times 140 - 15 = 265$

Parts b, a)

g) $2^3 + \dots + 20^3 = \sum_{r=2}^{20} r^3 = \sum_{r=1}^{20} r^3 - 1^3$
 $= \frac{20^2 \times 21^2}{4} - 1 = 44099$

$$Q2. \sum_{r=5}^{10} r^2 = \sum_{r=1}^{10} r^2 - \sum_{r=1}^4 r^2$$

Set to 1 and remove r you don't need i.e 1, 2, 3, 4 as want 5, 6, 7, 8, 9, 10

$$= \frac{10 \times 11 \times 21}{6} - \frac{4 \times 5 \times 9}{6} = 355$$

$$Q3. \sum_{r=1}^{10} (r^2 + 2r + 1) = \sum_{r=1}^{10} r^2 + \sum_{r=1}^{10} 2r + \sum_{r=1}^{10} 1$$

split

$$= \sum_{r=1}^{10} r^2 + 2 \sum_{r=1}^{10} r + \sum_{r=1}^{10} 1$$

$\downarrow$ $1+1+\dots+1$
10 lot's

$$= \frac{10 \times 11 \times 21}{6} + 2 \times \frac{10 \times 11}{2} + 10$$

$$= 505$$

$$Q4. \sum_{r=8}^{80} r^3 = \sum_{r=1}^{80} r^3 - \sum_{r=1}^7 r^3 = \frac{80^2 \times 81^2}{4} - \frac{7^2 \times 8^2}{4}$$

$$= 10496816$$

Q5. $\sum_{r=1}^{10} (r+1)^2 = \sum_{r=1}^{10} (r^2 + 2r + 1) = 505$
 by Q3.

$$\sum_{r=2}^{11} r^2 = \sum_{r=1}^{11} r^2 - 1^2 = \frac{11 \times 12 \times 23}{6} - 1$$

$$= 505 \text{ so yes}$$

As change index

$$\sum_{r=1}^{10} (r+1)^2 \quad \text{sub } k = r+1$$

$$\sum_{k=1+1}^{10+1} k^2 = \sum_{k=2}^{11} k^2 = \sum_{r=2}^{11} r^2$$

Change back as
 swapping variables
 has no change

Competency skills

Q6. for $1+2+\dots+n$ starting $a=1$,
 difference $d=1$

so $\sum_{r=1}^n r = \frac{1}{2} \times n \times (2a + (n-1)d)$

$$= \frac{1}{2} n (2 + n - 1) = \frac{n(n+1)}{2}$$

Q7. $u_r = r(3r-1) = 3r^2 - r$

$$\sum_{r=1}^n u_r = \sum_{r=1}^n (3r^2 - r) = 3 \sum_{r=1}^n r^2 - \sum_{r=1}^n r$$

$$= 3 \times \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} (2n+1 - 1)$$

$$= \frac{2n^2(n+1)}{2} = n^2(n+1)$$

Q8. a) $(1 \times 1) + (2 \times 3) + \dots + n(2n-1)$

so $u_r = r(2r-1)$

$$\text{so } \sum_{r=1}^n u_r = \sum_{r=1}^n r(2r-1)$$

$$= 2 \sum_{r=1}^n r^2 - \sum_{r=1}^n r$$

$$= 2 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$$

$$\begin{aligned}
 &= \frac{1}{6} (2n(n+1)(2n+1) - 3n(n+1)) \\
 &= \frac{1}{6} (n(n+1)(2(2n+1) - 3)) \\
 &= \frac{1}{6} n(n+1)(4n-1)
 \end{aligned}$$

Q9. $u_r = r^2$ need to sum

$$n^2 < u_r < 9n^2 \text{ and } (3n)^2$$

$$(n+1)^2 < u_r \leq (3n-1)^2$$

so

$$\sum_{r=n+1}^{3n-1} r^2 = \sum_{r=1}^{3n-1} r^2 - \sum_{r=1}^n r^2$$

replace $n \rightarrow 3n-1$

$$= \frac{(3n-1)(3n)(6n-1)}{6} - \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{1}{6} (n(3(3n-1)(6n-1) - (n+1)(2n+1)))$$

$$= \frac{n}{6} (3(18n^2 - 9n + 1) - (2n^2 + 3n + 1))$$

$$= \frac{n}{6} (54n^2 - 27n + 3 - 2n^2 - 3n - 1)$$

$$= \frac{n}{6} (52n^2 - 30n + 2)$$

$$= \frac{n}{3} (26n^2 - 15n + 1)$$

Q10. $(1 \times 2 \times 3) + (2 \times 3 \times 4) + \dots + (18 \times 19 \times 20)$

$$u_r = r(r+1)(r+2)$$

$$= r(r^2 + 3r + 2) = r^3 + 3r^2 + 2r$$

from $r=1$ to 18

$$\text{So } \sum_{r=1}^{18} u_r = \sum_{r=1}^{18} (r^3 + 3r^2 + 2r)$$

$$= \sum_{r=1}^{18} r^3 + 3 \sum_{r=1}^{18} r^2 + 2 \sum_{r=1}^{18} r$$

$$= \frac{18^2 \times 19^2}{4} + 3 \times \frac{18 \times 19 \times 37}{6} + 2 \times \frac{18 \times 19}{2}$$

$$= 29241 + 6327 + 342 = 35910$$

Q11. let $\sum_{r=1}^n r = 200$ and solve for n

$$\frac{n(n+1)}{2} = 200 \quad n^2 + n = 400$$

$$n^2 + n - 400 = 0$$

$$n = \frac{-1 \pm \sqrt{1^2 + 1600}}{2}$$

$$= \frac{-1 \pm \sqrt{1601}}{2} \quad \text{so no}$$

← not an integer solution

Advanced skills

Q12. a) so sequence from top to bottom is

$$\begin{array}{ccccccc} 100 & , & 81 & , & \dots & , & 4 & & 1 \\ 10 \times 10 & , & 9 \times 9 & , & \dots & , & 2 \times 2 & & 1 \times 1 \\ 10^2 & , & 9^2 & , & \dots & , & 2^2 & & 1^2 \end{array}$$

$$\text{So } \sum_{r=1}^{10} r^2 = \frac{10 \times 11 \times 21}{6} = 385$$

b) $4n^2$ up to n^2

$$\text{So } \sum_{r=n}^{2n} r^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n-1} r^2$$

$$= \frac{2n \times (2n+1) \times (4n+1)}{6} - \frac{(n-1)n(2n-1)}{6}$$

$$= \frac{1}{6} n (2(8n^2 + 6n + 1) - (2n^2 - 3n + 1))$$

$$= \frac{1}{6} n (16n^2 + 12n + 2 - 2n^2 + 3n - 1)$$

$$= \frac{1}{6} n (14n^2 + 15n + 1)$$

Q13. To prove the square sum

use $(r-1)^3 = r^3 - 2r^2 + 3r - 1$

so $r^2 = \frac{1}{3} (r^3 - (r-1)^3 + 3r - 1)$

so $\sum_{r=1}^n r^2 = \frac{1}{3} \left(\sum_{r=1}^n r^3 - \sum_{r=1}^n (r-1)^3 + 3 \sum_{r=1}^n r - \sum_{r=1}^n 1 \right)$

$= \frac{1}{3} \left(\sum_{r=1}^n (r^3 - (r-1)^3) + \frac{3n(n+1)}{2} - n \right)$

$= \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right)$

$= \frac{1}{3} \left(\frac{2n^3 + 3n^2 + 3n - 2n}{2} \right)$

$= \frac{1}{6} (2n^3 + 3n^2 + n)$

$= \frac{1}{6} n (2n^2 + 3n + 1)$

$= \frac{1}{6} n (2n+1) (n+1)$

$= \frac{n(n+1)(2n+1)}{6}$

~~$1^3 - 0^3 +$~~
 ~~$2^3 - 1^3 +$~~
 ~~$\vdots$~~
 ~~$n^3 - (n-1)^3$~~
 $= n^3 - 0^3$
 $= n^3$
 like
 method of
 differences